CTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html40/loose.dtd">

Thank you for filling out the API registration survey

Using the programmatic interface for Declination

The calculator provides an easy way for you to get results in HTML, JSON, XML, or CSV programmatically. To use the programmatic interface, you just have to url-encode the form parameters into a GET request including the result format you want.

A maximum of 50 connections/second from all the users is allowed at any time. If this is exceeded, the back-end calculator stops taking requests for 10 seconds. Bulk calculations should be requested serially and not in parallel. If the requests start returning with errors then your script should go to sleep for 5-10 minutes before trying the next request.

EXAMPLE:

An XML document with the magnetic declination in Boulder today:
https://www.ngdc.noaa.gov/geomag-web/calculators/calculateDeclination?lat1=40&lon1=-105.25&key=zNEw7&resultFormat=xml

BASE URL: https://www.ngdc.noaa.gov/geomag-web/calculators/calculateDeclination

PARAMETERS

Parameter	Required	Default	Description
key	yes		The access key is **zNEw7**. We will send any changes to the API information to your registered email address.
lat1	yes		decimal degrees or degrees minutes seconds: -90.0 to 90.0
lon1	yes		decimal degrees or degrees minutes seconds: -180.0 to 180.0
model		WMM	which magnetic reference model to use: 'WMM', 'WMMHR', or 'IGRF' For 'EMM' use base url: https://emmcalc.geomag.info with paramater magneticComponent=d
startYear		current year	year part of calculation date; WMM: 2024-2029, WMMHR 2024-2029, IGRF: 1590-2029
startMonth		current month	month part of calculation date: 1 - 12
startDay		current day	day part of calculation date: 1 - 31
resultFormat		html	format of calculation results: 'html', 'csv', 'xml', 'json'

Using the programmatic interface for U.S. Historic

The calculator programmatically provides results in HTML, XML, or CSV. To use the programmatic interface, url-encode the form parameters into a GET request and specify the desired result format.

A maximum of 50 connections/second is allowed at any given time. If exceeded, the calculator ceases requests for 10 seconds. Bulk calculations should be requested serially and not in parallel. If the requests start returning with errors then your script should go to sleep for 5-10 minutes before trying the next request.

EXAMPLE:

An XML document with U.S. Historic magnetic field values in Boulder for 2000 through 2010:
https://www.ngdc.noaa.gov/geomag-web/calculators/calculateUshistoric?lat1=40&lon1=-105.25&startYear=2000&endYear=2010&key=yKc9F&resultFormat=xml

BASE URL: https://www.ngdc.noaa.gov/geomag-web/calculators/calculateUshistoricc

PARAMETERS:

Parameter	Required	Default	Description
key	yes		The access key is **yKc9F**. We will send any changes to the API information to your registered email address.
lat1	yes		decimal degrees or degrees minutes seconds: 25.0 to 52.0
lon1	yes		decimal degrees or degrees minutes seconds: -130.0 to -65.0
startYear		current year	start of calculation interval: 1750-2030
endYear		current year	end of calculation interval: 1750-2030
resultFormat		html	format of calculation results: 'html', 'csv', or 'xml'

Using the programmatic interface for Magnetic Field

The calculator provides an easy way for you to get results in HTML, JSON, XML, or CSV programmatically. To use the programmatic interface, you just have to url-encode the form parameters into a GET request including the result format you want.

A maximum of 50 connections/second from all the users is allowed at any time. If this is exceeded, the back-end calculator stops taking requests for 10 seconds. Bulk calculations should be requested serially and not in parallel. If the requests start returning with errors then your script should go to sleep for 5-10 minutes before trying the next request.

EXAMPLE:

An XML document with magnetic field values in Boulder once a year since today's date in 2010 to the present using the World Magnetic Model
https://www.ngdc.noaa.gov/geomag-web/calculators/calculateIgrfwmm?lat1=40&lon1=-105.25&model=WMM&startYear=2025&endYear=2029&key=EAU2y&resultFormat=xml

BASE URL: https://www.ngdc.noaa.gov/geomag-web/calculators/calculateIgrfwmm

PARAMETERS

Parameter	Required	Default	Description
key	yes		The access key is **EAU2y**. We will send any changes to the API information to your registered email address.
lat1	yes		decimal degrees or degrees minutes seconds: -90.0 to 90.0
lon1	yes		decimal degrees or degrees minutes seconds: -180.0 to 180.0
elevation		0.0	elevation of the location: -10 km to 10,000 km
elevationUnits		K	units of elevation parameter: 'K', 'M', or 'F'
coordinateSystem		M	'D' (GPS) or 'M' (Mean Sea Level)
model		WMM	which magnetic reference model to use: 'WMM', 'WMMHR', or 'IGRF' For 'EMM' use base url: https://emmcalc.geomag.info
startYear		current year	start year of calculation interval: WMM: 2024-2029, WMMHR: 2024-2029, IGRF: 1590-2029
startMonth		current month	start month of calculation interval: 1 - 12
startDay		current day	start day of calculation interval: 1 - 31

Parameter	Required	Default	Description
endYear		current year	end year of calculation interval: WMM: 2024-2029, WMMHR: 2024-2029, IGRF: 1590-2029
endMonth		current month	end month of calculation interval: 1 - 12
endDay		current day	end day of calculation interval: 1 - 31
dateStepSize		1.0	size of steps through calculation interval in decimal years: >= .1
resultFormat		html	format of calculation results: 'html', 'csv', 'json', or 'xml'

Using the programmatic interface for magnetic field component grid

The calculator provides an easy way for you to get results in JSON, XML, or CSV programmatically. To use the programmatic interface, you just have to url-encode the form parameters into a GET request including the result format you want.

A maximum of 50 connections/second from all the users is allowed at any time. If this is exceeded, the back-end calculator stops taking requests for 10 seconds. Bulk calculations should be requested serially and not in parallel. If the requests start returning with errors then your script should go to sleep for 5-10 minutes before trying the next request.

EXAMPLE:

An XML document with a grid of inclination values near Boulder today with data points every tenth of a degree in latitude and longitude:
https://www.ngdc.noaa.gov/geomag-web/calculators/calculateIgrfgrid?lat1=39.8&lat2=40.3&lon1=-105.5&lon2=-105&latStepSize=0.1&lonStepSize=0.1&magneticComponent=i&key=gFE5W

BASE URL:https://www.ngdc.noaa.gov/geomag-web/calculators/calculateIgrfgrid

PARAMETERS

Parameter	Required	Default	Description
key	yes		The access key is **gFE5W**. We will send any changes to the API information to your registered email address.
lat1	yes		southern-most latitude bound in decimal degrees or degrees minutes seconds: -90.0 to 90.0
lat2	yes		northern-most latitude bound in decimal degrees or degrees minutes seconds: -90.0 to 90.0
latStepSize		1.0	size of steps through latitude interval in degrees: >= .1
lon1	yes		western-most longitude bound in decimal degrees or degrees minutes seconds: -180.0 to 180.0
lon2	yes		eastern-most longitude bound in decimal degrees or degrees minutes seconds: -180.0 to 180.0
lonStepSize		1.0	size of steps through longitude interval in degrees: >= .1
elevation		0.0	elevation of the location: -10 km to 10,000 km
elevationUnits		K	units of elevation parameter: 'K', 'M', or 'F'
coordinateSystem		M	'D' (GPS) or 'M' (Mean Sea Level)

Parameter	Required	Default	Description
magneticComponent	yes		magnetic component to include in result grid, one of: d = Declination i = Inclination x = North component y = East component z = Vertical intensity h = Horizontal intensity f = Total intensity
model		WMM	which magnetic reference model to use: 'WMM', 'WMMHR', or 'IGRF'
startYear		current year	start year of calculation interval: WMM: 2024-2029, WMMHR: 2024-2029, IGRF: 1590-2029
startMonth		current month	start month of calculation interval: 1 - 12
startDay		current day	start day of calculation interval: 1 - 31
endYear		current year	end year of calculation interval: WMM: 2024-2029, WMMHR: 2024-2029, IGRF: 1590-2029
endMonth		current month	end month of calculation interval: 1 - 12
endDay		current day	end day of calculation interval: 1 - 31
dateStepSize		1.0	size of steps through calculation interval in decimal years: >= .1
resultFormat		xml	format of calculation results: 'xml', 'json', or 'csv'